PRODUCER AND CONSUMER PROBLEM USING SEMAPHORES

CODE & OUTPUT :

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Activities   Terminal                    Apr 19  14:27
                          tejini@localhost:~/pc
[tejini@localhost pc]$ cat producer.c
#include<stdio.h>
#include<pthread.h>
#include<semaphore.h>
#define BUFFER_SIZE 5
int buffer[BUFFER_SIZE];
int in =0;
int out = 0;
sem_t empty;
sem_t full;
sem_t mutex;
void *producer(void *arg)
{
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                item=i;
                sem_wait(&empty);
                sem_wait(&mutex);
                buffer[in]=item;
                printf("Producer: %d\n",item);
                in = (in + 1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&full);
        }
}
void *consumer(void *arg){
        int item;
        for(int i=0;i<BUFFER_SIZE;i++){
                sem_wait(&full);
                sem_wait(&mutex);
                item=buffer[out];
                printf("Consumed: %d\n",item);
                out = (out+1)%BUFFER_SIZE;
                sem_post(&mutex);
                sem_post(&empty);
        }
}
int main()
{
        pthread_t p, c;
        sem_init(&empty,0,BUFFER_SIZE);
        sem_init(&full,0,0);
```

```
        pthread_create(&p,NULL,producer,NULL);
        pthread_create(&c,NULL,consumer,NULL);
        pthread_join(p, NULL);
        pthread_join(c, NULL);
        sem_destroy(&empty);
        sem_destroy(&full);
        sem_destroy(&mutex);
        return 0;
}
[tejini@localhost pc]$ gcc producer.c
[tejini@localhost pc]$ ./a.out
Producer: 0
Producer: 1
Producer: 2
Producer: 3
Producer: 4
Consumed: 0
Consumed: 1
Consumed: 2
Consumed: 3
Consumed: 4
[tejini@localhost pc]$ =
```